



Notes

UNDERSTANDING CHILDREN'S IDEAS AND ALTERNATE CONCEPTIONS

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2.1 INTRODUCTION

The following is the discussion between a teacher and her class 4th learners, while discussing an activity from the lesson on shadow from the NCERT Textbook for Class 4, Our Wondrous World (The World Around Us):

The teacher asked, “Does everyone have a shadow?” Children remained quiet. Then she shared a visual of a child standing in the sun, with the child’s shadow on the ground. After this promptly a few responded as follows:

- Yes, everyone has a shadow
- No... only animals have shadows.
- Plants don’t have. It is because animals move and plants don’t.
- Small things make small shadows; big things make big shadows.

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The above responses of the children seem incorrect. Some children were associating shadows to movement. Their explanations were not scientifically correct. You must have come across responses like these every day. However, hard you try to address these, you may have found that it is difficult to change their reasoning. Let me give a few more instances of children's scientifically incorrect responses of physical phenomena around us.

- Plants obtain their food from the soil.
- Hollow objects float and heavy objects sink.
- Seeds grow because they eat soil.
- Ice melts faster in the Sunlight.
- Plants are not alive because they can't walk.
- Objects that feel hotter are at a higher temperature.
- Clouds are made up of cotton.
- When clothes are kept for drying, water disappears.

These are not specific to any culture or country. It is across children in primary classes. Sometimes, it is not limited to young children even adults have such scientifically incorrect conceptions. For instance, you must come across even some adults who have this notion that the earth is flat. Technically, we read in the school that the earth is spherical but many adults are unable to comprehend that how can people live on something that is spherical like a ball. If unaddressed, their incorrect conceptions remain with them throughout their lives. Thus, it is important for you to understand what children think so that their thinking becomes a starting point for your pedagogies in the classroom. In this chapter, we would try to comprehend about how children learn concepts from the perspective of constructivist theorists (John Dewey, and Vygotsky). Further, we would elaborate upon Driver's conception of alternative frameworks, children's ideas and strategies to identify and address children's preconceptions/ misconceptions.



2.2 LEARNING OUTCOMES

After completion of this chapter, the learner

- comprehends how children think, learn concepts and make sense of the world around them;
- observes how children learn concepts in science and social science;

- elaborates upon the theory of Piaget and its implication for concept development for primary school children;
- explains the theory of Vygotsky and its implication for concept development for primary school children;
- distinguishes between preconception and misconception;
- analyses and uses appropriate strategies for identifying, probing and addressing children's ideas in the classroom;
- evaluates and links effective teaching and curriculum design to children's learning.

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You must have observed that children in your class have responses to age-appropriate questions. For instance, on asking examples of living things, children would say a train, the moon and the sun and “plants” as non-living. Have you ever wondered, why is it so? To understand this, let's see how children in the primary classes learn concepts. As you have earlier read that children are active beings. They are not passive receivers of information. The developmental view of learning emphasised experiential and age-appropriate learning.

Children learn through many ways mainly by being active and engaging in concrete experiences. For instance, whether an object floats or sinks, children learn by doing it themselves. By engaging in hands-on experiences, they learn to observe, compare, classify, manipulate objects (variables). For instance, they may initially observe that a piece of paper or paper boat floats but after a while it may sink. Thus, they learn new concepts (which may be incorrect or incomplete in some cases) by building on prior experiences and knowledge.

You must have also observed that language and interaction also play an important role in children's learning. Imagine, a child living near a sea shore comes across the word ‘desert’ in a discussion. Just by mere introduction of the word, the child tends to explore more about it by talking, asking questions from peers and adults. This kind of exploration would lead to learning.

Children also learn by making connections, new linkages to their existing knowledge. They begin with mental models, say animals, which become more and more complex like pet animals, domestic animals, wild animals, animals living on land, and water and so on. These interconnected webs become more complex over a period of time.

Children also learn by trial and error. For instance, a child may think that plants need sunlight to grow but after experimenting with different plants they

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would learn that plants also need water, fertilizer, appropriate soil and weather conditions for growth. Their learning is ongoing, gradual but not linear. Let's take an example of a Class 3 teacher, where she performs an activity by putting an inverted glass on a burning candle. On asking why the candle extinguished after a few seconds of covering it with a glass tumbler, children would come up with various responses like – the wick of the candle touched the tumbler, or the wick turned black, or probably it is a windy or a rainy day. The eight-year-old children would not be able to link it to air or oxygen as a supporter of combustion. It shows that learning does not happen immediately or instantaneously, but it is a gradual process and takes time. Only with repeated exposure and meaningful activities using visuals (videos, pictures, charts or models), and stories (real life examples) children refine their knowledge.

2.3.1 Piaget's Theory of Cognitive Development

As you have already read about Jean Piaget's theory of conceptual development, you must be aware of the stages of cognitive development as proposed by him. The primary school child is either in the Preoperational Stage (2–7 years) or Concrete Operational Stage (7–11 years) depending on her or his age. Children's thinking and learning is deeply influenced by the stage they are in. Let's summarize the cognitive characteristics of both the stages.

Cognitive characteristic of primary school children (age 2 to 7 years) in the pre-operational stage

- Children's thinking is egocentric i.e. they understand their own perspective. They have difficulty in comprehending situations from other perspectives. For instance, a girl playing hide and seek, closes her eyes and thinks that as she can't see others likewise others can't see her too.
- Children's thinking is intuitive and imaginative which is based on appearances, not logic. For instance, a six year old would say that it is raining because clouds are crying.
- Children have a tendency to believe that non-living things have feelings or intentions. It is known as animism. You must have come across children conversing, feeding, bathing their dolls.
- Children struggle with conservation of number, length, and quantity (volume). If there are two equal rows of coins having 5 coins each, and in one row the coins are spread out, children in this stage think that the longer row has a greater number of coins. Children tend to focus on length, not on numbers.
- Children have a limited understanding of cause-effect relationships focusing only on visible or immediate factors. They make connections

on the basis of coincidence, not casualty. Their thinking is perceptually dominated. A child may say that, “It rained today because I forgot to bring my umbrella.”

Thus, according to Piaget, children in the pre-operational stage learn by their personal experiences. They learn concrete concepts, things which could be seen. In other words, it can be said that their learning is visual. Thus, it is recommended that as a teacher, you should use pictures, real objects, familiar examples and stories while teaching these children. You should allow them to explore freely, encourage talking and drawing.

Cognitive characteristic of primary school children (age 7 to 10 years) in the concrete-operational stage:

- Children develop key cognitive abilities namely, conservation, classification, seriation and reversibility at this stage.

Conservation – Ability to understand that quantity remains the same even when appearance changes.

Classification – Ability of grouping objects on the basis of some common properties (colour, shape, size, material or any other).

Seriation – Ability to arrange objects in an order on the basis of some property like length, or mass; such as from shortest to tallest, or lightest to heaviest.

Reversibility – An understanding that things can return to their original state by reversing the action; water can be turned into ice and vice-versa.

- Children are able to reason logically with concrete objects and events. They are able to make mental connections like if $P > Q$ and $Q > R$ then $P > R$

Thus, according to Piaget, children in the concrete-operational stage learn through concrete examples. They are able to compare, group, and classify objects and events. Their learning is more organized, systematic, rule-based and they can comprehend the relationship between concepts. They start making sense of the world by seeking explanations and asking “why” and “how” questions. As a teacher, teaching children in this age-group, you should use hands-on activities, experiments, and group work, connecting to real-life experiences thus, encouraging discussion and reasoning.

2.3.2 Vygotsky's Socio-Cultural Theory and Conceptual Development

The socio-cultural theory of Vygotsky recommended that learning is a social



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process and happens best through interaction with more knowledgeable people - teachers, parents, peers, or community members. He proposed that when children interact socially through talks, conversations, tasks or play they learn new ways of thinking. His idea of ZPD, 'Zone of Proximal Development' gained appreciation, which is defined as "the gap between what a child can do independently and what they can do with help". Vygotsky recommended that the most effective learning happens within this zone, where a child is challenged but simultaneously supported. For instance, a 9-year-old may not understand the concept of shadow, however with support of her or his peers, and teachers, the child may grasp it. This support could be visual aids, questions, cues, and prompts that helps children in learning the concept. Vygotsky called this temporary support as scaffolding, which is gradually withdrawn once children understand the said concept. During scaffolding, language plays an important role in development of thinking and concept formation. Children are talking to each other and adults; they engage in self-talk; and through these processes, children refine their ideas. Thus, Vygotsky believed that language is a tool for thinking.

You as a teacher of primary school children must encourage them to talk through their thinking, ask 'why' and 'how' questions, use concept-specific vocabulary, use step-by-step explanations for concepts, peer assistance, and scaffolding.



CHECK YOUR PROGRESS 2.1

1. Which of the following statement is NOT true about learning:
 - i) Learning happens with repeated exposure and meaningful activities
 - ii) Learning happens immediately or instantaneously with stories
 - iii) Learning happens by making connections, new linkages to the existing knowledge
 - iv) Language and interaction play an important role in children's learning
2. According to Piaget's theory of cognitive development, the primary school children are in which of the following stages
 - i) Preoperational and concrete operational stage
 - ii) Sensory-motor and preoperational stage
 - iii) Concrete operational and formal operational stage
 - iv) Formal operational and sensory-motor stage
3. Which of the following is NOT a cognitive characteristic of primary school children in the pre-operational stage:



- | | |
|----------------|---------------------|
| i) Egocentrism | ii) Animism |
| iii) Intuitive | v) Logical Thinking |
4. Which of the following cognitive ability is NOT associated with the children in the concrete-operational stage:
- | | |
|---------------------|-------------------|
| i) Conservation | ii) Egocentrism |
| iii) Classification | iv) Reversibility |
5. Which of the following concept is NOT proposed by Vygotsky in his Socio-Cultural Theory:
- | | |
|--------------------|----------------------------------|
| i) Scaffolding | ii) Zone of Proximal Development |
| iii) Accommodation | iv) Self-Talk |

Answers:

1. ii), 2. i), 3. iv), 4. ii), 5. iii)

2.4 ALTERNATIVE FRAMEWORK

You must have come across various instances wherein children's responses are scientifically incorrect. For instance, 'the Earth is flat'; 'the Earth is round like a plate'; 'Water disappears from the wet clothes, when put to dry under the sun' and so on. These are not random errors but coherent, experience-based explanations. Rosalind Driver, a constructivist science education researcher along with her colleagues, studied how children form their own explanations of natural phenomena. She labelled children's ideas as "alternative framework". An alternative framework refers to "a coherent set of ideas or explanations that children (or learners) form to make sense of the world, which differs from the accepted scientific or academic understanding".

Consider the following conversation of a class 4th teacher with her students. The teacher showed an image in which a shadow is formed of a child standing in the sun. She asked, "What do you see in the picture?" After a few responses like "child, sun, plants," a child responded 'shadow'. Immediately the teacher nodded "yes", and asked a further question, "What do you know about shadows?" The students gave a variety of responses as listed below:

- It is dark.
- It is black in colour.
- It is always black in colour.
- It follows me.

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- Yes! It always follows me when I walk in the sun. I think it's like my friend.
- It runs after me! When I jump, it jumps too! It's like it's copying me.
- It is like my twin.
- It sits down with me; it does the same thing as me if I move my hand up it also lifts its hand.
- Yes! One time I saw myself on the wall and I waved, and it waved back.

Then the teacher said, "Ok. At what time of the day do you see shadows?" The students replied:

- Only in the daytime! At night it's too dark for shadows to be seen.
- At night, when the light is turned on, I can see a shadow.
- No ... it is not there at night.
- No, not at night. Because we can't see anything in the dark.
- No, I can't see anything in the dark.

The Teacher further asked, "How are shadows formed?" The students' responses were:

- I see it when there is light, may be the light makes it.
- When I stand in the sun, it comes behind me. I think the ground makes it.
- The shadow remains still even if I move the light.
- When I move, it runs behind me! Sometimes it tries to catch me!
- When I go near it, it becomes small, I think it runs away from me.
- When I try to touch it my hand goes through! It's like it's not really there.
- I tried to step on it but it ran away.
- It doesn't move when I stand still.

The teacher continued, "You seem to know a lot about shadows. But, does the size of your shadow remain the same? Can you make shadows of objects?" (Referring to the activity in the textbook, NCERT Class 4) The following were the responses from students:

- No, sometimes it's bigger than me and sometimes smaller.

- When I jump, it gets bigger!
- Tiny things make small shadows like their size.

You must be having similar interactions with your students. You must have observed that these responses are not random. Children have logically consistent frameworks in their mind, which are based on their personal experiences, prior knowledge, observations, and sometimes, their intuitive reasoning. These provide an internal logic that guides how children interpret new information. These are alternative frameworks, also called naïve theories.

Since, 1980s a lot of research has been conducted on children's ideas and features of their ideas are identified as follows:

Ideas are personal: An idea of one child is different from the other child. From the above interaction, the child thought that the ground creates the shadow. She/he is not realizing that a shadow is formed when the way of light is blocked. Children's ideas are based on their experiences of what they see, hear, or do in their daily lives, not on underlying mechanisms.

Ideas are unique: A child believes that the shadow is her or his friend or twin. Other children may not believe the same. Children do not realise that shadows can't be formed without light. Many believed that a shadow is a real, touchable object that just disappears.

Ideas are stable: Many children thought shadows are always of the same size. They didn't realise that the size of a shadow is dependent on the light source with respect to the image. However hard, one may try to convince children, their ideas are deeply rooted and resistant to change. Even if their ideas are scientifically incorrect, they often have an internal logic based on reasoning.

Ideas are incomplete or partially correct: Children may understand some parts of a concept correctly. They have undifferentiated concepts. For instance, a child might believe that "heavier objects fall faster than lighter ones". She or he does not seem to differentiate between a number of concepts in this situation like – mass and weight of the object, acceleration due to gravity, force acting on the object, friction forces, resistance of air, motion and the concept of vacuum.

Ideas are contradictory: Children may hold multiple, even conflicting ideas at the same time, and shift between them depending on the context. For instance, they may say that clouds are made up of cotton in one context and in another, they may agree that clouds are full of water, depending on the question asked.

As a teacher, you need to encourage children to participate in discussions, interactions and hands-on learning. You should respect their thinking as meaningful, not just 'wrong'. Their ideas should be starting points for the teaching and learning process. You need to plan your lessons to help learners to restructure their understanding through active engagement in activities. Thus,

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such constructivist approaches in teaching assist in exploration of children's thinking and focus not on superficial changes but rather deep-rooted conceptual changes.

2.4.1 Children's Preconceptions vs Misconceptions

Preconceptions are the ideas formed before formal instruction in the classroom. It is often incomplete or partially correct, vague, and based on everyday life experiences, observations, or cultural/environmental influences. These show a limited understanding of children. For instance, heavier objects sink in water or the Sun rotates around the Earth. On the contrary, misconceptions are scientifically incorrect understanding formed after formal instruction. It may develop due to misinterpretation, poor teaching, or oversimplified explanations. These can hinder further learning. For instance, after a lesson, a child believes that salt and sugar disappear when it is dissolved in water.

Thus, knowing whether an idea is a preconception or a misconception helps you decide how to address it in the classroom. You should build your classroom instruction on what children know. Your pedagogical strategies should include concrete materials, visuals, stories, hands on-activities, real life examples, discussion, talking, questioning and explaining.

2.4.2 Sources of Alternate Conceptions

You must be wondering what causes alternate conceptions among children. A few prominent sources of alternate conceptions are listed in the diagram given on the next page below:

Everyday Experiences: Children use their everyday life experiences to form ideas, which may be limited, causing misconceptions. For instance, when children see the sun rising and setting they form a misconception that the sun moves across the sky.

Usage of Language: Some common phrases or explanations used by adults may lead to confusion. For instance, phrases like “a kilogram of milk”, “the sun rises in the east” or “plants eat food” can lead children to form incorrect ideas.

Unclear Instruction: Sometimes, teachers teach abstract concepts, which may be difficult to understand for a primary school child leading to partial or incorrect understandings. For instance, the concept of the water cycle is too abstract for primary school children. Many children believe that water droplets on the outer surface of a tumbler having cold water have actually leaked from inside.

Influence of media: Children absorb ideas from cartoons, stories, animation from comic books, television or other platforms like YouTube, which are often fantasy or fiction solely for entertainment. For instance, ‘people walking on

clouds', 'cars as living beings', 'the earth is round like a ball and we live inside it'. Such unrealistic or magical ideas create alternate conceptions.

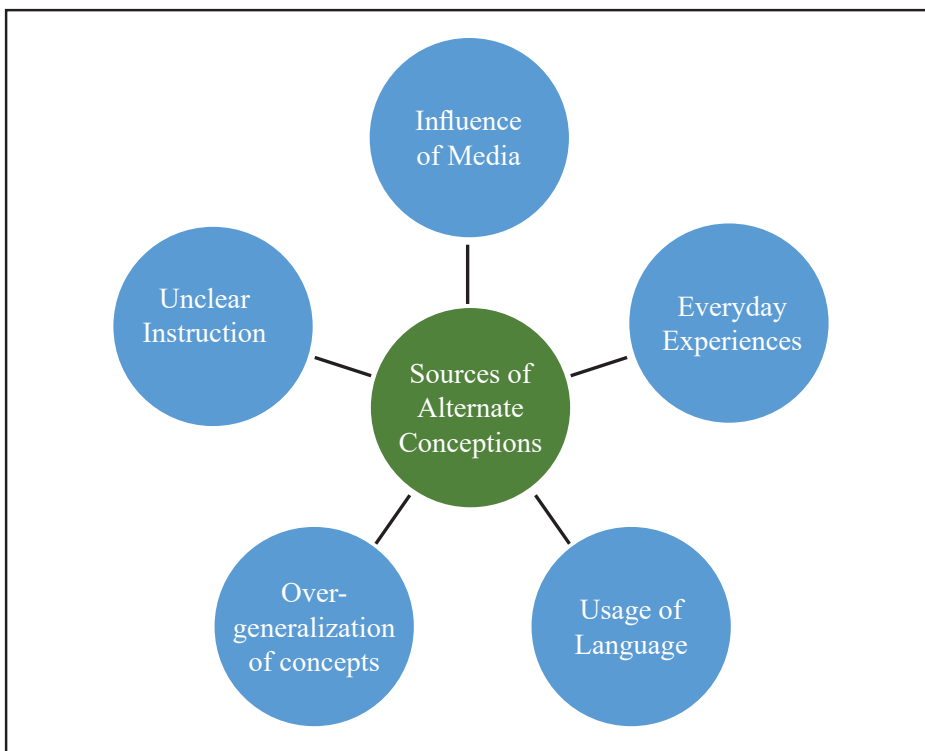


Fig. 2.1: Sources of Alternate Conceptions

Over Generalization of Concepts: Children sometimes apply a familiar idea to all situations, even when it doesn't fit. For instance, children believe that potatoes and onions don't grow under the soil but on plants, as they have seen fruits and vegetables growing on plants.

Thus, if you are able to identify the source of alternate conception then it would be easier to address it through appropriate planning and interventions. Therefore, it is advised that you should use questions, observation, and discussion to uncover and address these ideas.



CHECK YOUR PROGRESS 2.2

1. Which of the following is NOT a characteristic feature of children's ideas:
 - i) Ideas are personal
 - ii) Ideas are resistant to change
 - iii) Ideas are correct
 - iv) Ideas are stable

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2. A coherent set of ideas or explanations that learners form to make sense of the world, which differs from the accepted understanding is known as:
 - i) Alternative Framework
 - ii) Pre-conception
 - iii) Misconceptions
 - iv) Scientifically Correct Conception
3. To address children's ideas, the pedagogical strategies of the teacher should not focus on which one of the following?
 - i) Concrete materials and Hands on activities
 - ii) Visuals and discussions
 - iii) Stories and real-life examples
 - iv) Scientific correct explanation

Answers:

1. iii), 2. i), 3. iv)

2.5 IDENTIFYING CHILDREN'S IDEAS IN THE CLASSROOM

By now you have understood that, children have preconceptions or alternate conceptions regarding concepts, which are incorrect or partially correct. Therefore, identification of children's ideas in the classroom is a crucial step in helping children build accurate concepts. These are many ways through which you can identify their alternate conceptions:

Open-Ended Questions in The Class Discussion: Try to engage children in discussion and interaction and ask open ended questions using 'how' or 'why'. Such questions encourage explanation and reflection. As a teacher you should avoid recall based questions and questions that require responses as 'yes' or 'no'. For instance, pose questions like 'how do plants get their food?'; 'how do plants make their own food?' Such questions prompt children for reasons and explanations, which help us in finding out about children's reasoning, and their ideas.

Drawing and Modelling: Children's drawings reveal their understanding much more than their explanations. For instance, asking children to draw a rabbit facing to the right and a lion facing to the left. Their drawings let us understand not only their sense of directions (right and left) but also their understanding of relative sizes of both the animals and the perspectives (front view, top view)

through which they are drawing these animals. You should encourage children to draw or build models. Such visuals often provide much more details as compared to their verbal explanations.

Hands-on Activities: You should plan and organize hands-on activities for children. Encourage them to “what will happen?” and then conduct the activity and observe and record it. For instance, a number of objects can be provided to children for finding out “what sinks and what floats in water?”. They can maintain a table like the one shown below:

Table 2.1: Predict- Observe - Explain Table

S. No.	Object	Prediction	Observation	Explanation
1	A bottle cap	Float	Float	It is light so it floats
2	Marble	Sink	Sink	It is heavy so it sinks
3	Eraser	Float	Sink	It is small so it sinks
4				

You should observe children for how they behave and what they talk during activities and play. It would help you in identifying their thinking pattern, explanation and curiosity.

Concept Mapping: It provides a linkage between broad, cross-functional concepts. These use boxes and connecting lines to show the connection between concepts and sub-concepts.

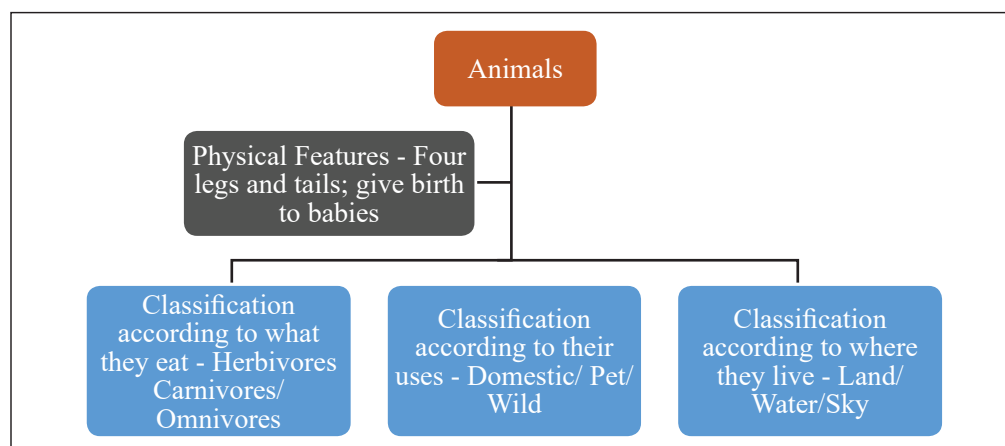


Fig. 2.2: Concept Map on Animals Class 2

When making concept maps children make connections, which let us know how they organize their knowledge and understand relationships between different concepts/ sub-concepts/ ideas. As a teacher, you should encourage them to make concept maps on any concept and ask them to group, sort or link those.

Interviews: You could engage children in informal conversation or conduct interviews either in dividedly or as a group. Such conversation should be



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friendly and natural, not a test. This kind of interaction helps us in revealing deeper ideas of children. In the previous section, the conversation of children in the shadows was described. Using prompts, cues and visuals would facilitate the conversation.

Storytelling: You must be narrating stories to children in your primary classes. You must have noticed that children enjoy listening to stories. For instance, 'Journey of a River', 'Story of a Seed', 'How the Earth Shook' (Titles of the books by National Book Trust, Delhi) are some of the examples of stories related to The World Around Us. Children could be engaged in role play to make it more participative. After storytelling, you could ask questions like "What would you say if you were the river?"; "Tell me a story about how seeds grow". Such questions encourage children to think and give us an idea about their understanding of sequence of events, explanation of the events and their alternative frameworks.

Thus, using various strategies a teacher could identify children's misconceptions. You should remember that a child friendly, comfortable, and safe environment would encourage children to open up and share their responses. An environment without fear and where they know that they are not judged and labelled help children to open up. You need to value all responses, especially incorrect ones, as those are the ones which would give you an insight about how children make sense of the world. Thereby you need to adapt your planning and teaching strategies for addressing their misconceptions.

2.6 STRATEGIES TO ADDRESS ALTERNATE CONCEPTIONS

You must have understood that alternate conceptions are a natural part of learning. However, these need to be identified and addressed carefully. The following strategies could be used in the classroom:

Using Discrepant Events:

The discrepant events are "unexpected or surprising phenomena that challenges children's existing understanding of concepts". For instance, an egg sinks in water, but it floats when salt is added to it. Such events encourage curiosity and critical thinking among children as they try to figure out the 'why' of unexpected outcomes. Children rethink their ideas when reality contradicts them.

Using Hands-On Experiences:

As you know, children in the primary classes learn the best by doing things themselves. You should give them materials and objects to touch, feel, see, taste, and explore ideas directly. The hands-on activities help them in building or modifying their alternative framework. For instance, providing magnets and iron items to children, help them in learning about attraction and repulsion.

Using Stories, and Visuals:

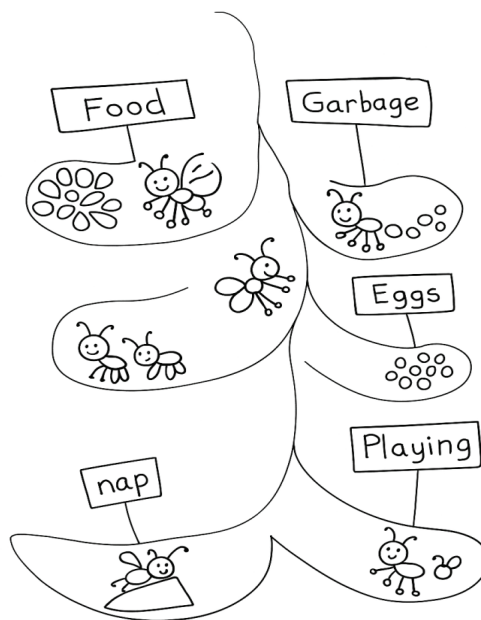
At times, stories and pictures or visuals simplify abstract ideas thus making it easier for children to relate or develop analogies. For instance, telling a story about a raindrop would help children in understanding the explanation of processes of evaporation and condensation thereby making sense of the water cycle.

Using Discussion and Interaction:

As recommended by Vygotsky, children should be encouraged to participate in discussions and interaction in small groups and as an entire class, wherein they talk with peers and adults. This type of scaffolding within the zone of proximal development support children in organizing their thoughts and ideas.

Using Drawings and Models:

A teacher of Class 1, took children to a park and asked them to observe insects and draw. A few children carefully observed ants and drew their pictures in their notebook. The following is a picture drawn by a 5 year old child. The image tells us a lot of things like - Ants are happily playing and moving in a line; ants are near garbage, and sugar; ants are taking a nap. The image also shows ant's eggs. Typically the size of ant eggs is about 0.5 mm and it is practically not possible to see these with the naked eye. Secondly it would not be possible for children to see ants sleeping, as their rest period is very small and secondly, they rest in their created chambers. Thus, it is inferred from this sample that it is a mixture of their imagination and observation. Such tasks reveal children's deep-rooted misconceptions and give you a platform to discuss and address their alternative conceptions.



As a teacher of primary classes, you must have understood that you do not have to label children's ideas as "wrong", instead, explore and address those. You need to start with what the child believes and use multiple strategies to develop their alternate conceptions. One should remember that a single strategy does not work for everyone. Always use a mix of strategies. This would keep your classroom environment lively and foster creativity among children.

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2.7 CONCLUSION: ROLE OF THE REFLECTIVE TEACHER

Reflective teachers constantly think about their planning, pedagogy, student responses, and classroom process for making learning better for children. You should not simply follow a fixed lesson plan but adapt your pedagogies based on how children think and respond. It begins with observation of children in the classroom. You need to pay attention to minute details of what children say, write, draw or make, which would help you in identifying their thinking patterns. It is then followed by asking probing and open-ended questions. Such questions don't test their recall of factual knowledge but rather assist you in identifying children's alternative framework. For instance, instead of asking, "Do plants need sunlight?" you should ask, "Why do you think that plants need sunlight?" In the process you need to constantly ask yourself "How was your class today?"; "Did children like it?"; "Did they understand the concept?" "Why didn't they understand it?"; "What did I need to do differently so that they learn the concept?" You should plan your lesson in such a way that you address their alternative conceptions. In other words, you should start your lessons with children's existing ideas and build upon them.

If children believe that a rainbow is caused by flying a helicopter in the sky then respect that idea. Don't discourage the idea and say that it's silly. Instead plan an activity to show splitting of light and let them visualise how rainbows are formed. In this scenario, you are not only demonstrating the concept but also creating a space wherein children are free to express their ideas. It should be noted that interaction with children is not optional but rather it is essential for children's conceptual growth.

Thus, children are more likely to share their ideas. You should treat misconceptions as natural and use these as a starting point for learning. You need to adopt various strategies to identify, explore and address their misconceptions in the classroom. Working with children's ideas in the classroom, you engage with action research. You act as a researcher of children's thinking wherein you need to continue observation, reflection, and adaptation of your teaching strategies. In a nutshell effective teaching involves eliciting, respecting, and then restructuring children's ideas toward scientific understanding.

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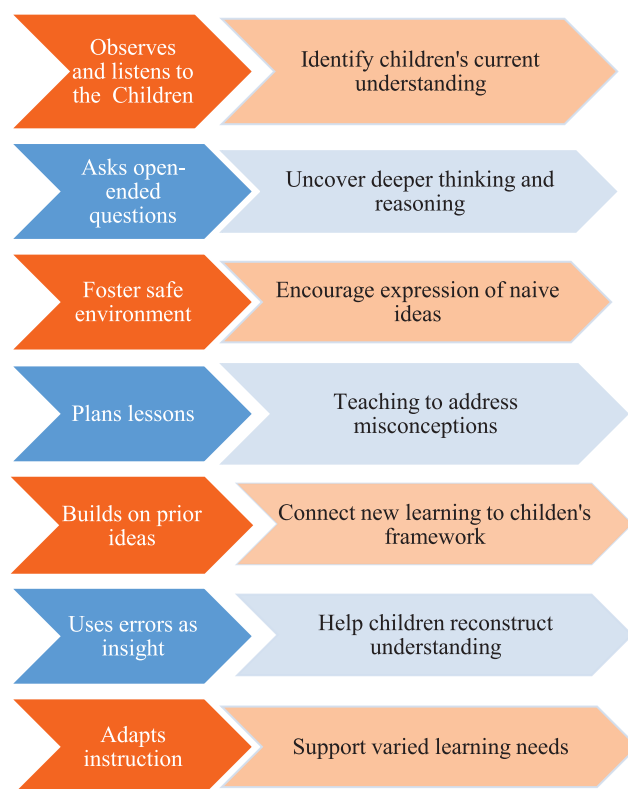


Fig. 2.4: A Reflective Teacher's Action and its Impact

2.8 LET US SUM UP

- Children's ideas are valuable starting points for teaching. They are not "mistakes," but windows into how children think.
- Children actively construct their own understanding of the world, using prior knowledge, experiences, and social interaction.
- Alternate conceptions are not random errors. These are logical in nature and based on children's experiences and prior knowledge.
- Constructivist teaching should address alternate conceptions to help children reconstruct their understanding through active learning.
- Inspired by Piaget, the constructivist pedagogy uses activities that challenge children's current ideas, which create conflict and thus motivates children to revise their thinking.
- Inspired by Vygotsky, constructivism emphasizes talking, listening, and reasoning with peers and teachers to clarify understanding. Concepts are co-constructed when adults and peers engage children in meaningful dialogue and activity.

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- The primary school children need both hands-on experience and social guidance to develop understanding of concepts.
- Constructivist teachers observe children's ideas continuously, and see misconceptions not as errors to "correct," but as stepping stones toward deeper learning. With appropriate support, children can revise, reconstruct and modify their ideas and develop scientific meaningful understanding.
- A reflective teacher does not aim to "fix" children's wrong answers but to understand their logic and guide them patiently towards scientifically accurate, meaningful learning.

2.9 END EXERCISES

1. Explain using examples from your classroom how you would identify and address children's misconceptions?
2. How do the theories of Piaget and Vygotsky guide you in understanding children's learning? Elaborate using examples.
3. Explain the characteristics of children's ideas in the primary classrooms.
4. Distinguish between pre-conception and misconception.

2.10 REFERENCES

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